

NetWatch

Packet Sniffer & Anomaly Detector

REVA Semester Project

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1. Abstract

NetWatch is an integrated Intrusion Detection and Prevention System (IDS/IPS) that captures live network traffic using Scapy, analyses packets for suspicious behaviour, and presents results on a real-time web dashboard. The system detects SYN floods, port scans, stealth TCP scans, web attack signatures via Deep Packet Inspection (DPI), and unknown protocols. High-severity alerts trigger automatic firewall blocking on Windows and Linux. Passive intelligence features extract DNS queries and TTL-based OS fingerprints. Optional threat intelligence enrichment uses AbuseIPDB and VirusTotal APIs. Captured traffic can be exported as PCAP for Wireshark analysis, and forensic PDF reports map detections to the MITRE ATT&CK framework.

2. Introduction

Network security monitoring is essential for identifying malware activity and unauthorized access attempts. Traditional packet sniffers provide visibility but lack automated anomaly detection. NetWatch combines live packet capture, heuristic intrusion detection, active prevention, and a modern web interface into a single platform suitable for educational and lab environments.

2.1 Objectives

- Capture and display live network packets using Scapy.
- Detect anomalies indicative of scanning, flooding, and web attacks.
- Provide real-time visualisation via a web dashboard with WebSocket updates.
- Export PCAP files for deep inspection in Wireshark.
- Optionally block high-severity attackers at the OS firewall (IPS mode).
- Generate forensic PDF reports with MITRE ATT&CK technique mapping.

3. System Architecture

NetWatch uses a three-layer architecture: (1) the network interface and OS firewall, (2) the Python backend (Flask, Scapy, detection engines), and (3) the browser dashboard. A producer-consumer queue decouples packet capture from analysis: the Scapy sniffer thread only enqueues packets, while worker threads parse, detect, and emit results.

Components:

- Sniffer thread: Scapy sniff() with fast enqueue-only producer.
- Worker pool (2 threads): parse packets, update stats, run detection.
- Sweeper thread: expires firewall block rules after 10 minutes.
- Flask-SocketIO server: serves dashboard and pushes live events.
- Background threads: IPS mitigation and threat-intel API lookups.

4. Technologies Used

- Python 3 - core language
- Scapy - packet capture and protocol parsing
- Flask + Flask-SocketIO - web server and real-time WebSocket communication
- Chart.js - protocol distribution and packet rate charts

- Tailwind CSS + Lucide icons - dashboard UI
- fpdf2 - forensic PDF report generation
- python-dotenv - environment variable configuration
- Wireshark - offline PCAP analysis (external tool)
- AbuseIPDB / VirusTotal APIs - optional IP reputation enrichment

5. Anomaly Detection Rules

The following heuristics are implemented in `check_anomalies()`:

Detection	Threshold	Severity	MITRE ID
SYN Flood	>20 SYNs in 5s	HIGH	T1498
Port Scan	>15 ports in 10s	MEDIUM	T1046
NULL Scan	TCP flags = 0	HIGH	T1046
XMAS Scan	FIN+PSH+URG	HIGH	T1046
Brute Force	>=15 auth ports in 30s	MEDIUM	T1110
DNS Tunneling	Query >=80 chars	MEDIUM	T1071.004
DGA DNS Activity	>50 unique in 60s	MEDIUM	T1568
C2 Beacon	>=8 periodic to public IP	HIGH	T1071
Suspicious Payload	LOLBin signature	HIGH	T1105
DPI (SQLi/XSS/etc.)	Signature match	HIGH	T1071
Unknown Protocol	Non TCP/UDP/ICMP	LOW	-

5b. REVA Malware and Unauthorized Access Detections

NetWatch extends baseline IDS rules with REVA-focused detections for malware activity and unauthorized access. Alerts are tagged with categories (Reconnaissance, Unauthorized Access, Malware Activity, Web Exploitation, Denial of Service, Protocol Anomaly) for dashboard filtering and forensic PDF grouping.

- Brute Force Attempt: repeated connections to FTP, SSH, Telnet, RDP, SMB, or WinRM ports.
- DNS Tunneling: unusually long DNS query names (>=80 characters).
- DGA DNS Activity: burst of unique DNS queries from a single source (>50 in 60s).
- C2 Beacon: periodic, fixed-size outbound traffic to public destinations.
- Suspicious Payload: cleartext LOLBin and malware transfer signatures (PowerShell, certutil, etc.).

Limitations: brute force detection is packet-level (not login failure events); DNS rules are behavioral heuristics (CDN and AV update storms may false-positive); C2 beaconing requires public destinations; payload inspection cannot read HTTPS.

6. Intrusion Prevention (IPS)

When a HIGH-severity alert fires, NetWatch adds an OS firewall rule blocking the source IP for 600 seconds. On Windows, `netsh advfirewall` is used; on Linux, `iptables DROP` rules are applied. A background sweeper thread removes expired blocks every 15 seconds. MEDIUM and LOW alerts generate dashboard notifications only.

7. Passive Intelligence

- DNS extraction: query names decoded from DNSQR layer.
- TTL OS fingerprint: TTL <=64 Linux/Unix, <=128 Windows, else network device.
- Displayed in the intelligence rail and included in forensic PDF reports.

8. Threat Intelligence Integration

After HIGH or MEDIUM alerts on public IP addresses, NetWatch asynchronously queries AbuseIPDB (abuse confidence score, country, ISP) and VirusTotal (malicious engine count). Results are cached for 10 minutes and shown in the IP Threat Intel panel. Private LAN addresses are skipped. API keys are loaded from a .env file.

9. Web Dashboard

The dashboard provides a full-width live packet feed with sticky headers, an intelligence rail with statistics, charts, alerts, fingerprints, DNS queries, and threat intel. Control buttons (Start, Stop, Export PCAP, Report) use a unified segmented control with Lucide icons. Protocol colours are consistent across charts and table badges. Anomaly alerts display REVA category badges (Reconnaissance, Unauthorized Access, Malware Activity, etc.) with a client-side category filter.

10. Wireshark Integration

NetWatch does not embed Wireshark. Instead, users export captured packets as a PCAP file (capture.pcap) and open it in Wireshark for protocol-level analysis, filtering, and verification of detected anomalies.

11. Installation and Usage

- Install dependencies: `pip install -r requirements.txt`
- Copy .env.example to .env and configure keys (optional).
- Run as Administrator (Windows) or sudo (Linux): `python app.py`
- Open `http://127.0.0.1:5000`, click Start Capture.
- View live packets, alerts, and intelligence panels.
- Export PCAP for Wireshark or click Report for forensic PDF.

11b. Verification and Testing

Two helper scripts in scripts/ validate REVA detections. Offline verification runs synthetic Scapy packets through the detection engine without starting the web server: `python scripts/verify_reva_detections.py` (expect all checks PASSED). Live verification starts app.py, begins capture, and runs demo traffic such as `python scripts/demo_reva_traffic.py dns-tunnel or brute --target <lab-ip>`.

- Offline: `verify_reva_detections.py` - brute force, DNS, payload, beacon, categories.
- Live: `demo_reva_traffic.py` - dashboard alerts with category badges and filter.
- Forensic PDF: Report button - Detections by Category plus MITRE mapping.
- Wireshark: export PCAP and filter by alert source IP to confirm traffic.

Note: C2 Beacon and Suspicious Payload are HIGH severity and may trigger IPS auto-block. Run app.py as Administrator for capture and firewall rules. Forensic reports may show Host: Windows 10 on Windows 11 due to Python platform.release() compatibility reporting.

12. Limitations

- Captures on default network interface only.
- Heuristic detection may produce false positives.
- DPI cannot inspect encrypted HTTPS payloads.
- Firewall blocking affects inbound traffic to this host only.
- Threat intel APIs require public IPs and valid API keys.

13. Conclusion

NetWatch successfully fulfils the project requirement for a packet sniffer and anomaly detector using Scapy and Wireshark. It provides real-time monitoring, multiple detection techniques, optional active prevention, passive intelligence, external threat enrichment, and MITRE-mapped forensic reporting in an accessible web interface suitable for cybersecurity education and demonstration.

References

- Scapy documentation - <https://scapy.net>
- MITRE ATT&CK Framework - <https://attack.mitre.org>
- AbuseIPDB API - <https://docs.abuseipdb.com>
- VirusTotal API - <https://developers.virustotal.com>
- Flask-SocketIO - <https://flask-socketio.readthedocs.io>